LESSON 5.3 | Lesson notes

Debugging with gdb and valgrind

Overview

Compile with the -g option so the debugger knows your source lines. gdb lets you run the program, set a breakpoint with break and step through the code with next, printing variables with print. When the program crashes, the backtrace command shows which functions were running. Valgrind runs the program and reports invalid reads and writes, use of uninitialised values and memory that was never freed. Run valgrind on every program you write while you are learning; most pointer bugs show up immediately.

Key points

- Part of: Pointer Pitfalls and Debugging
- About 20 minutes
- Try the example yourself, then check it

How to use this handout

1. Read the overview first, then work through the key points in order.
2. Write down anything you are unsure about and ask in the course forum.
3. Use page 2 to plan your practice and keep notes.

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Practice checklist

- [ ] Read: Debugging with gdb and valgrind
- [ ] Try the example from the lesson
- [ ] Explain the main idea in your own words
- [ ] Answer the related questions
- [ ] Ask about anything unclear in the forum

My notes